

NOISEMAP · DESIGN PRECIS

NoiseMap design precis

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DOCUMENT CONTROL

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REVISION HISTORY

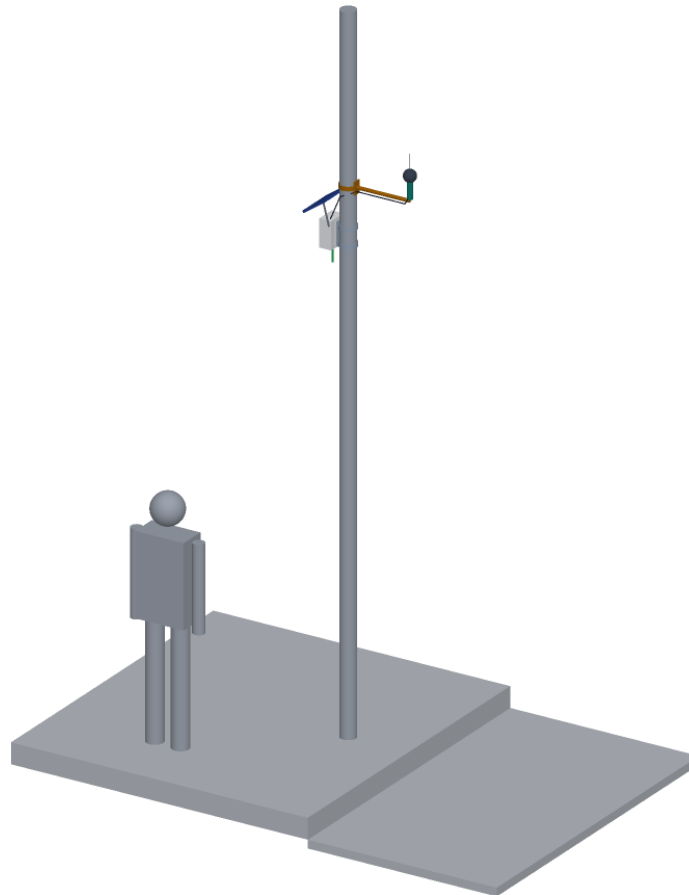
Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, safety, media)
0.1	2026-09-25	Amish Chadha	Initial scaffold

NoiseMap design precis

NoiseMap is a clamp-on street pole noise node. A digital MEMS microphone about 4 m above the ground feeds a small processor in the microphone head, which turns the sound into A- and C-weighted levels once a second and discards the audio within milliseconds. Only levels cross the cable to a standard FieldNode core, which sends a 22-byte record every 15 minutes over LoRaWAN. First-order numbers suggest the node draws about 21 mW, runs about 30 days without sun and measures from about 35 to 115 dBA, but the parts cost of about \$171 is over the \$100 budget.

NoiseMap

CONCEPT, NOT FOR FABRICATION



Grey: Street pole, 5 m, Sidewalk and road, Person, 1.75 m for scale

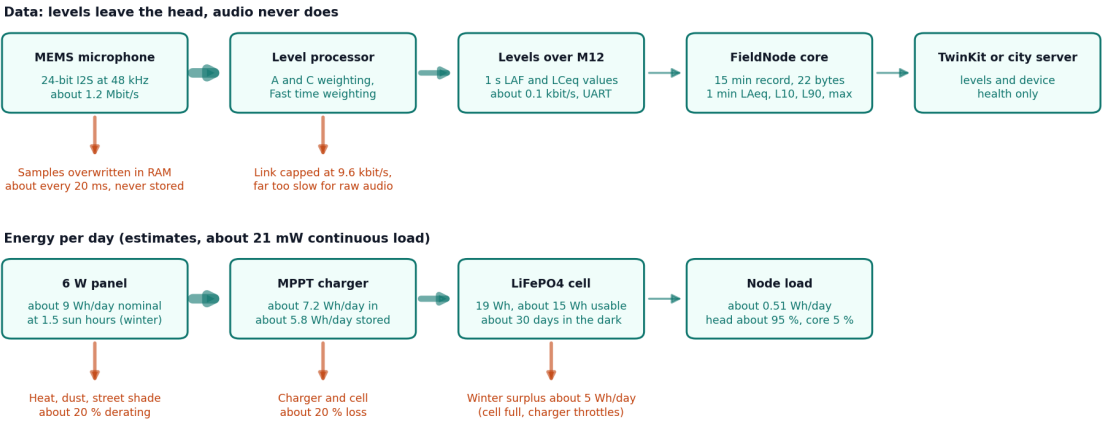
■ How it works

1. **Sense.** The MEMS microphone, port facing up inside a foam windscreen, streams 24-bit samples over I²S at 48 kHz to the level processor.
2. **Weight and integrate.** The processor applies A and C frequency weighting as digital filters, squares and averages the result, and computes each second the A-weighted equivalent level (LAeq,1s), the Fast time-weighted maximum and minimum (LAFmax, LAFmin) and the C-weighted equivalent level (LCeq,1s). Samples live in a small RAM buffer that is overwritten about every 20 ms.
3. **Hand over levels only.** Once a second the head sends about 12 bytes of levels to the FieldNode core over a 9,600 baud UART in the M12 cable. The head has no flash storage for data and no radio.

4. **Summarize.** The FieldNode core builds a 15-minute record: fifteen 1-minute LAeq values, the 15-minute LAeq, LAFmax and LAFmin, L10 and L90 (from the 900 one-second values), LCEq and a status byte, 22 bytes in all.
5. **Send.** The core sends each record over LoRaWAN to the lab's TwinKit gateway, a city server or The Things Network, and stores it in flash if the link is down. Night indicators (Lnight, Lden) are computed on the server from the 15-minute records.

NoiseMap: data and energy flow

CONCEPT, NOT FOR FABRICATION. Values are estimates to be checked at TRL 3.



■ **Main components**

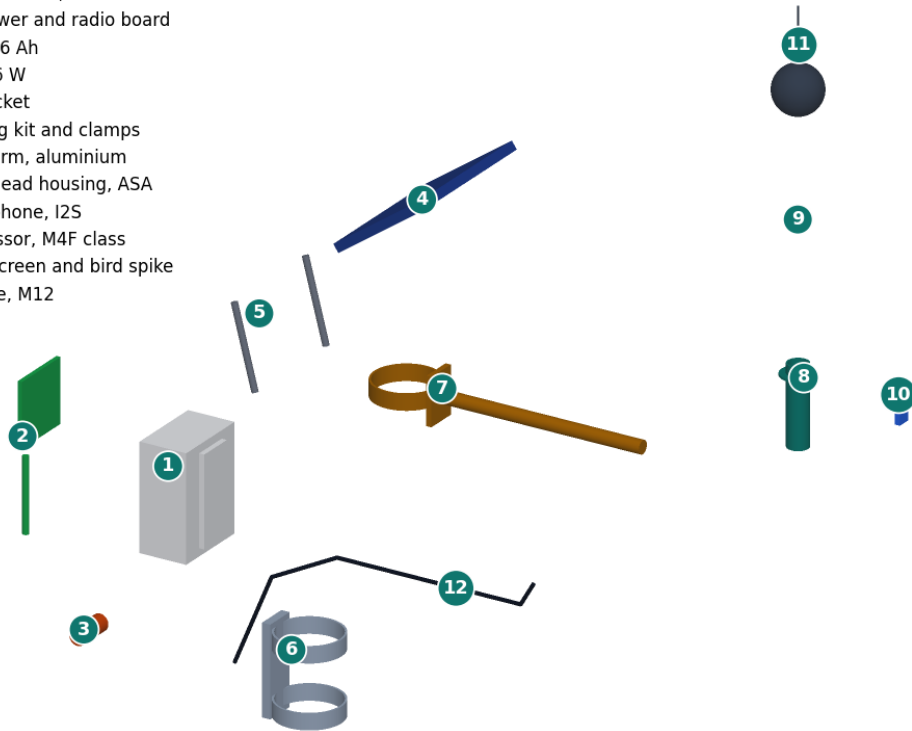
Table 1. Main components. Numbers match the exploded view (Figure 3) and `bom/bom.csv`.

#	Component	Proposed choice	Notes
1	FieldNode enclosure	IP65 polycarbonate box, about 200 x 150 x 90 mm, vent and gland	From FieldNode, unchanged
2	FieldNode power and radio board	MPPT charger with cold-charge lockout, STM32WL-class LoRaWAN module, SPI flash, M12 ports	From FieldNode, unchanged
3	LiFePO4 cell	One 3.2 V 6 Ah cell	FieldNode standard is enough
4	Solar panel	6 W, about 290 x 200 mm, doubling as a hood	FieldNode standard is enough
5	Panel tilt bracket	Flat-bar posts, about 35° tilt	From FieldNode
6	Pole mounting kit	Back plate and band clamps for 60 to 140 mm poles	Larger clamps than FieldNode's 40 to 60 mm kit
7	Microphone arm	25 mm aluminum tube, about 400 mm reach toward the street	Keeps the microphone about 0.45 m from the pole face
8	Microphone head housing	3D-printed ASA tube, 40 mm OD x 150 mm, drip skirt, acoustic membrane	Proposed, awaiting Amish
9	MEMS microphone	ICS-43434 class I ² S, 65 dB SNR, 120 dB SPL overload (TDK InvenSense)	Listed as end of life; alternative proposed, awaiting Amish
10	Level processor	Cortex-M4F class low-power board (STM32L4 class)	Proposed, awaiting Amish
11	Windscreen and bird spike	90 mm open-cell foam ball, bored to fit the head	Replace about yearly (estimate)
12	Sensor cable	M12 5-pin, about 1.5 m, 3.3 V and UART	FieldNode standard pinout

NoiseMap: exploded view

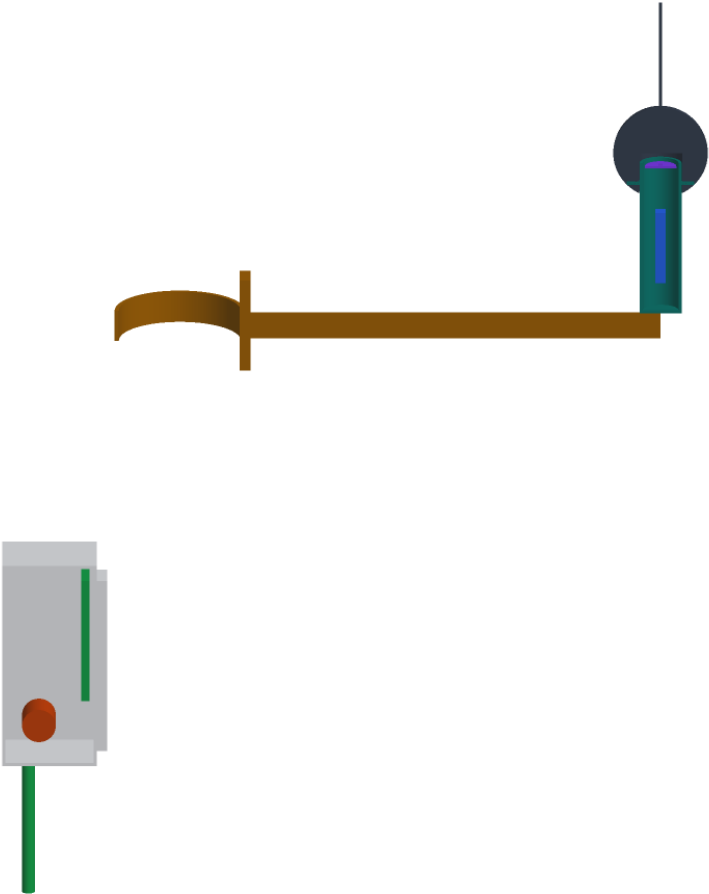
CONCEPT, NOT FOR FABRICATION

- 1 FieldNode enclosure, IP65
- 2 FieldNode power and radio board
- 3 LiFePO4 cell, 6 Ah
- 4 Solar panel, 6 W
- 5 Panel tilt bracket
- 6 Pole mounting kit and clamps
- 7 Microphone arm, aluminium
- 8 Microphone head housing, ASA
- 9 MEMS microphone, I2S
- 10 Level processor, M4F class
- 11 Foam windscreen and bird spike
- 12 Sensor cable, M12



NoiseMap: cutaway

CONCEPT, NOT FOR FABRICATION



■ First-order numbers

All values are estimates for concept review and will be checked at TRL 3.

Table 2. First-order numbers.

Quantity	Estimate	Basis and assumptions	Requirement
Microphone self-noise	about 29 dBA	94 dB SPL reference minus 65 dB SNR (datasheet class value)	
Lower measuring limit	about 35 dBA	Self-noise 6 dB below the signal adds about 1 dB; correctable by subtracting the noise floor	R4 at risk below 35 dBA
Upper measuring limit	about 115 dBA	120 dB SPL acoustic overload point, 5 dB margin for peaks	R4 met
Raw audio rate	about 1.2 Mbit/s, held only in RAM	48 kHz x 24 bits	R1
Level data over the cable	about 0.1 kbit/s	About 12 bytes per second including framing	R1 met by design
Uplink record	22 bytes per 15 min, 96 uplinks/day	15 x 1-byte 1-minute LAeq (0.5 dB steps), 6 statistics, 1 status	R2 met
Radio airtime	about 0.25 s per uplink, about 24 s/day	SF9 at 125 kHz, 35 bytes with LoRaWAN overhead; The Things Network fair use allows 30 s of uplink per day per node (TTN)	R12 met at SF9 or faster; not met at SF10 without longer intervals
Microphone and processor power	about 18 mW	About 0.5 mA microphone and 5 mA processor at 3.3 V (typical class values, to be confirmed)	
Total load from the cell	about 21 mW, about 0.51 Wh/day	Head 18 mW, FieldNode core and radio about 1 mW, rails 90 % efficient	Within FieldNode's about 115 mW sensor allowance
Battery autonomy	about 30 days	15.4 Wh usable (19.2 Wh cell, 80 %)	R7 met
Winter solar harvest	about 5.8 Wh/day into the cell	6 W x 1.5 peak sun hours x 0.8 derating x 0.85 charger x 0.95 cell	R7 met (about 11 times the load)
Mass on the pole	about 2.3 kg	FieldNode about 1.7 kg, arm 0.25 kg, head and windscreen 0.15 kg, cable and larger clamps 0.2 kg	R10 met
Wind load on the panel	about 50 N	0.058 m ² , 35 m/s gust, force coefficient 1.2	R10, bracket to check
Parts cost	about \$171 (FieldNode core about \$119, NoiseMap parts about \$52)	Indicative prices, see <code>bom/bom.csv</code>	R13 not met

The numbers above are for the electronics only. The accuracy of the measurement depends on the housing, the membrane, the windscreen, reflections from the pole and nearby facades, and wind, none of which can be settled on paper.

■ Key design choices

- **Levels, never audio.** The only data that leaves the microphone head are levels once a second. The head has no data storage and no radio, and the firmware has no code path that writes or sends samples. Proposed, awaiting Amish.
- **Where privacy comes from.** The 9,600 baud cable link is far too slow for raw audio (about 1.2 Mbit/s), and the LoRaWAN link averages well under 100 bit/s under duty-cycle and fair-use limits. Neither alone rules out a heavily compressed speech codec, so the privacy guarantee rests on open, auditable head firmware with read-out protection and a published build hash. This is stated plainly so no one relies on a stronger claim. Proposed, awaiting Amish.
- **Processing in the head, not in FieldNode.** Keeping the audio next to the microphone means the M12 cable only ever carries levels. Proposed, awaiting Amish.
- **Standard FieldNode core.** At about 21 mW the node fits FieldNode's sensor allowance with its standard 6 W panel and one cell, so no power upgrade is needed. Proposed, awaiting Amish.
- **Microphone about 4 m above the ground, 0.45 m from the pole, pointing up.** This matches noise-mapping practice and reduces the pole reflection. Proposed, awaiting Amish.
- **A and C weighting.** LAeq and its statistics follow common practice; LCeq, and the difference LCeq minus LAeq, flags amplified bass from venues that A weighting under-reports. Proposed, awaiting Amish.
- **Microphone part.** The ICS-43434 is widely used (DNMS supports it) but TDK lists it as end of life. The Infineon IM72D128 is the alternative DNMS already supports. Proposed: design for either on a small adapter board, awaiting Amish.
- **Field calibration.** Each node is checked with a sound calibrator (94 dB at 1 kHz) at install and at each windscreen change, and compared with a class 1 reference meter at one site. Proposed, awaiting Amish.

■ Safety

Safety: Installing on a street pole is work at height next to traffic. Install only with the pole owner's permission, by trained crews, with fall protection and traffic management as local rules require. Keep clear of overhead power lines and street-light wiring, and never open a pole's electrical hatch.

Safety: The node contains a LiFePO4 cell of about 19 Wh. Fuse the cell, charge only within the cell maker's temperature limits (FieldNode's cold-charge lockout applies), and do not install a node with a swollen or damaged cell.

Safety: A falling part from 4 m can injure people below. Fit a secondary safety lanyard to the microphone arm and check the clamps after the first storm.

Safety: Do not test the node near its upper range with loudspeakers or calibrators at close range without hearing protection.

Privacy. No images, audio recordings or personal identifiers leave the device; only sound levels are stored or sent. Check local data protection law before any deployment, publish the head firmware and its build hash, and put a notice on the pole saying what is measured and where the design is documented.

Use of the data. NoiseMap readings are indicative. They are not certified measurements and must not be presented as legal evidence without a co-located certified instrument.

■ Open questions for TRL 3

- Measure the frequency response and noise floor of the chosen microphone in the printed head, with and without the acoustic membrane and windscreen.
- Check the low-frequency response below about 60 Hz, which limits LCeq for amplified bass.
- Decide how to flag wind- and rain-contaminated intervals without a wind sensor (for example, from the spread of 1-second levels or low-frequency energy), or add a small wind sensor.
- Estimate the error from reflections off the pole and nearby facades at 0.45 m.
- Confirm processor current at 48 kHz with A and C filtering; check whether a lower sample rate is acceptable.
- Choose the first co-design partner and street, and agree how residents and the city will share the data.

Concept media: [blueprint sheet](#), [interactive 3D model](#).